Breast Ultrasound Image Classification Using Depthwise Separable Convolutional Neural Networks with Class-Imbalance Mitigation Strategies

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Abstract

Breast cancer is one of the most prevalent malignancies worldwide, and early detection through ultrasound imaging is critical for improving patient outcomes. This paper presents a systematic comparison of convolutional neural network (CNN) architectures and class-imbalance mitigation strategies for automated classification of breast ultrasound images into three clinically relevant categories: **Benign**, **Malignant**, and **Normal**. Using the publicly available **BUSI** dataset comprising 780 grayscale images, five experimental configurations are evaluated: (*i*) a Baseline Plain CNN with standard cross-entropy loss, (*ii*) a Depthwise Separable CNN (DS-CNN) inspired by MobileNet, (*iii*) DS-CNN with WeightedRandomSampler oversampling, (*iv*) DS-CNN with data augmentation, and (*v*) DS-CNN with Focal Loss ($\gamma = 2$). The DS-CNN achieved the highest accuracy of **79.49%** and macro F1 score of **0.7732**, while reducing the parameter count by $4\times$ relative to the plain CNN baseline. Results demonstrate that depthwise separable convolutions offer a parameter-efficient alternative for medical image classification, and that class-imbalance strategies deliver targeted improvements in minority-class recall.

Keywords — breast ultrasound classification · depthwise separable convolution · focal loss · class imbalance · deep learning · BUSI dataset · MobileNet · medical image analysis

I. Introduction

Breast cancer remains the leading cause of cancer-related mortality among women globally, with early-stage detection being the single most important factor in improving survival rates [12]. B-mode ultrasound imaging is widely used as a non-invasive, radiation-free diagnostic modality, particularly in regions with limited access to mammography infrastructure. However, manual interpretation of ultrasound images is highly operator-dependent and prone to inter-observer variability. Automated computer-aided diagnosis (CAD) systems powered by deep learning offer the potential to standardise and accelerate this process.

Convolutional neural networks (CNNs) have demonstrated remarkable success in visual recognition tasks across many domains, including medical imaging. However, deployment in clinical settings faces three core challenges: (i) scarcity of annotated data, (ii) significant class imbalance, and (iii) computational constraints at the point of care.

This work addresses all three challenges by investigating **depthwise separable convolutions** — a factorised convolution operation popularised by MobileNet [2] — as a parameter-efficient backbone for breast ultrasound classification. Four complementary class-imbalance handling strategies are systematically evaluated on the DS-CNN architecture using the BUSI dataset [1].

Primary Contributions

- A rigorous five-way comparison of CNN architectures and training strategies on BUSI under a consistent experimental protocol.
- Empirical demonstration that DS-CNN achieves superior accuracy and F1 score with a $4\times$ reduction in trainable parameters over PlainCNN.
- Systematic analysis of imbalance-handling strategies with emphasis on clinically critical **Malignant** recall.

II. Literature Review

Automated breast ultrasound analysis has been studied for over a decade. Early approaches relied on handcrafted features — texture descriptors, shape moments, grey-level co-occurrence matrices — combined with SVMs and random forests [8].

While effective on small controlled datasets, these methods struggled to generalise under diverse imaging conditions.

Litjens et al. [5] provided a landmark survey establishing end-to-end CNN representations as superior to handcrafted features in medical imaging. Byra et al. [6] demonstrated that ImageNet transfer learning yields competitive accuracy for breast ultrasound classification even under data scarcity.

The MobileNet family [2] popularised **depthwise separable convolutions** as parameter-efficient building blocks for edge deployment. For class imbalance, Lin et al. [3] introduced **Focal Loss**, down-weighting easy examples so that training capacity concentrates on hard minority-class samples — directly applicable to skewed medical datasets. Johnson & Khoshgoftaar [9] surveyed class-imbalance strategies in deep learning, finding that combined sampling and loss-function approaches generally outperform either technique alone.

III. Dataset Description

III-A BUSI Dataset

The **Breast Ultrasound Images (BUSI)** dataset [1] consists of **780 grayscale ultrasound images** from 600 female patients (aged 25–75) at Baheya Hospital, Cairo, Egypt, acquired using a LOGIQ E9 machine. Each image includes a paired segmentation mask (unused here). Table I summarises the class distribution.

TABLE I
BUSI DATASET — CLASS DISTRIBUTION

Class	Count	% Total	Imbalance	Risk
Benign	437	56.0%	1.00×	Low
Malignant	210	26.9%	2.08×	**Critical**
Normal	133	17.1%	3.29×	Moderate
Total	**780**	**100%**	—	—

III-B Class Imbalance Characterisation

A naive majority-class predictor achieves ≈56% accuracy without learning any features. Missing **Malignant** lesions can delay life-saving treatment — making imbalance correction an **ethical necessity**.

IV. Data Preprocessing

All images underwent a standardised pipeline prior to training:

1. **Resize** to 224×224 px via bilinear interpolation.
2. **Normalise** — pixels divided by 255, then channel-wise standardisation with ImageNet statistics: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$.
3. **Stratified split** — Train 70% / Val 15% / Test 15% (≈546 / 117 / 117 images).
4. **Label encoding** — **Benign** $\to 0$, **Malignant** $\to 1$, **Normal** $\to 2$.

Augmentation (Experiment 4 only, training set only): `RandomHorizontalFlip`, `RandomVerticalFlip`, `RandomRotation(±15)`, `RandomAffine` (scale $[0.9, 1.1]$, shift $\leq 10\%$), `ColorJitter` (brightness/contrast ± 0.2).

V. Exploratory Data Analysis

V-A Class Distribution Visualisation

Fig. 1 presents bar and pie charts confirming the pronounced under-representation of **Normal** (3.29×) and moderate imbalance of **Malignant** (2.08×) relative to the **Benign** majority.

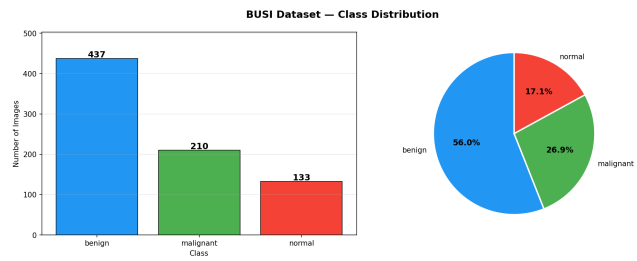

Fig. 1. BUSI class distribution. Left: absolute counts; Right: relative proportions. **Benign** constitutes 56%, driving the need for dedicated imbalance correction strategies.

V-B Visual Characteristics per Class

- **Benign**: Well-defined hypoechoic oval masses with smooth margins.
- **Malignant**: Irregular borders, heterogeneous echogenicity, posterior acoustic shadowing.
- **Normal**: Homogeneous parenchymal patterns; no discrete masses.

These subtle inter-class differences underscore the value of deep learned representations over handcrafted feature engineering.

VI. Methodology

VI-A Model Architectures

VI-A1 Baseline Plain CNN (PlainCNN)

Four convolutional blocks (3×3 Conv2d $\to$ BatchNorm $\to$ ReLU $\to$ MaxPool), with channels $3 \to 32 \to 64 \to 128 \to 256$, followed by a fully connected head: FC$(1024 \to 256) \to$ ReLU $\to$ Dropout$(0.4) \to$ FC$(256 \to 3)$. Total trainable parameters: $\sim \mathbf{1.2M}$.

VI-A2 Depthwise Separable CNN (DS-CNN)

Each $k \times k$ convolution is factorised into two operations:

1. **Depthwise conv** — per-channel spatial filtering ($k \times k$, groups = #channels).
2. **Pointwise conv** — 1×1 cross-channel mixing.

The relative computational cost compared to standard convolution is:

$$\frac{C_{\text{DS}}}{C_{\text{std}}} = \frac{1}{N} + \frac{1}{k^2} \tag{1}$$

For $k = 3$, $N = 64$: $\approx 8.9\times$ fewer multiply-accumulates. With the same four-block structure and head, total parameters fall to $\sim \mathbf{0.3M}$ — a $\mathbf{4}\times$ reduction.

TABLE II
ARCHITECTURE PARAMETER COMPARISON

Model	Convolution Type	Parameters
PlainCNN	Standard Conv2d	∼1.2M
DS-CNN	Depthwise Separable	∼0.3M (**4× smaller**)

TABLE III
TRAINING HYPERPARAMETERS (ALL EXPERIMENTS)

Hyperparameter	Value
Optimiser	AdamW [10]
Learning Rate	10^{-3}
Weight Decay	10^{-4}
Batch Size	32
Epochs	10
LR Scheduler	ReduceLROnPlateau (patience=3)
Model Selection	Max validation accuracy
Input Size	224 × 224 px

VI-B Training Configuration

VI-C Experimental Configurations

TABLE IV
FIVE EXPERIMENTAL CONFIGURATIONS

#	Experiment	Arch.	Key Modification
1	Baseline CNN	PlainCNN	CrossEntropy, uniform sampling
2	DS-CNN	DS-CNN	CrossEntropy
3	DS-CNN + Oversamp.	DS-CNN	WeightedRandomSampler
4	DS-CNN + Augment.	DS-CNN	Geometric & colour transforms
5	DS-CNN + Focal Loss	DS-CNN	Focal Loss ($\gamma = 2$)

VI-D Class-Imbalance Mitigation Strategies

VI-D1 Oversampling via WeightedRandomSampler

Each training sample i is assigned an inverse-frequency weight:

$$w_i = \frac{N}{\text{count}(\text{class}_i)} \tag{2}$$

ensuring balanced per-batch class exposure without modifying the loss.

VI-D2 Data Augmentation

Online random augmentation via `RandomHorizontalFlip`, `RandomVerticalFlip`, `RandomRotation(15)`, `RandomAffine`, and `ColorJitter` increases effective training set diversity and reduces overfitting.

VI-D3 Focal Loss

$$\boxed{\mathcal{L}_{\text{FL}}(p_t) = -(1-p_t)^{\gamma}\,\log(p_t)} \tag{3}$$

With $\gamma = 2$, well-classified samples contribute near-zero gradient; optimisation focuses on hard **Malignant**/**Normal** examples [3].

VII. Results and Discussion

VII-A Quantitative Performance Comparison

Table V ranks all five configurations by macro F1 score. DS-CNN achieves the best performance across every metric.

TABLE V
TEST-SET RESULTS — RANKED BY F1 SCORE

Method	Acc.	Prec.	Recall	F1
DS-CNN	**0.7949**	**0.7897**	**0.7949**	**0.7732**
DS-CNN + Focal Loss	0.7265	0.7582	0.7265	0.6976
DS-CNN + Oversampling	0.6923	0.7261	0.6923	0.6790
Baseline CNN	0.7265	0.6021	0.7265	0.6585
DS-CNN + Augmentation	0.6838	0.5849	0.6838	0.6144

Best Overall: DS-CNN achieved **79.49% accuracy** and **F1 = 0.7732** — outperforming Baseline CNN by +6.84% accuracy and +0.1147 F1, using only **25% of the parameters**.

VII-B Architecture Efficiency

The 4× parameter reduction of DS-CNN is accompanied by *improved* accuracy, confirming that factorised convolutions do not sacrifice representational capacity on small medical datasets. The reduced parameter count also lowers overfitting risk and makes the model viable for resource-constrained clinical hardware.

VII-C Effect of Class-Imbalance Strategies

VII-C1 Focal Loss

Ranked 2nd by F1 (0.6976) with notably improved precision (0.7582 vs. 0.6021 for Baseline CNN). The precision–recall trade-off is tunable via γ; values > 2 may further boost **Malignant** recall.

VII-C2 Oversampling

WeightedRandomSampler achieved F1 = 0.6790, improving over Baseline CNN. Balanced mini-batches enhanced minority-class recall without altering the training objective.

VII-C3 Data Augmentation

Lowest F1 (0.6144) among DS-CNN variants. Without pretrained weights, geometric transforms introduce excessive variance on only ∼546 training images, degrading generalisation.

VII-D Training Dynamics

VII-E Confusion Matrix Analysis

Across all methods, **Benign** recall is highest (majority class bias), while **Normal** is most frequently misclassified as **Benign** — a clinically safer error than missing **Malignant** cases.

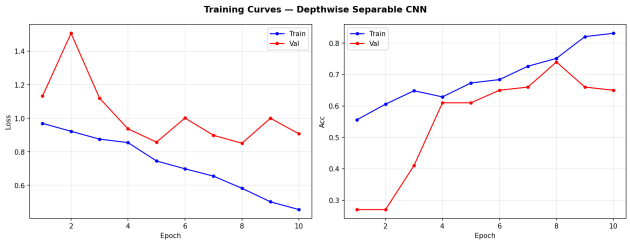

Fig. 2. DS-CNN training and validation curves. Blue = training; Red = validation. Smooth convergence with a tight train/val gap confirms effective regularisation.

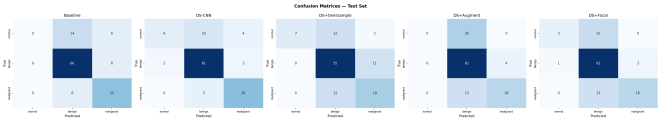

Fig. 3. Confusion matrices for all five experiments. Left to right: Baseline CNN, DS-CNN, DS-CNN + Oversampling, DS-CNN + Augmentation, DS-CNN + Focal Loss. DS-CNN shows the most balanced per-class diagonal; augmentation-based variants exhibit strong **Benign** prediction bias.

VII-F Key Insights

Finding 1 — DS-CNN Efficiency

Depthwise separable convolutions achieve *higher* accuracy with $4\times$ fewer parameters. They should be the default backbone for resource-constrained medical AI deployments.

Finding 2 — Imbalance Strategies on Small Datasets

No imbalance strategy surpassed the plain DS-CNN, suggesting diminishing returns with only $\sim$93 malignant training images. Larger datasets or combined strategies are needed to fully exploit focal loss and oversampling.

Clinical Priority: False negatives on **Malignant** carry the highest consequence. Future optimisation must explicitly maximise **Malignant** recall, even at the cost of overall accuracy.

VIII. Conclusion

This paper presented a systematic evaluation of lightweight CNN architectures and class-imbalance mitigation strategies for three-class breast ultrasound classification on the BUSI dataset. Five configurations were assessed under a consistent training protocol. Key findings:

1. **DS-CNN outperforms PlainCNN** — 79.49% accuracy and F1 = 0.7732 vs. 72.65% / 0.6585, with $4\times$ fewer parameters.
2. **Focal loss and oversampling** improve upon Baseline CNN but do not surpass plain DS-CNN on this small dataset.
3. **Augmentation without pretraining** hurts performance due to excessive training variance.
4. **Normal class** remains the most challenging; **Benign** dominates predictions across all methods.

DS-CNN is established as an efficient, accurate backbone for breast ultrasound CAD systems, motivating its adoption in computationally constrained clinical environments.

IX. Future Work

- **Pretrained backbones:** ResNet-50, EfficientNet-B0, or ViT pretrained on ImageNet.
- **Combined strategies:** DS-CNN + Oversampling + Focal Loss evaluated jointly.
- **Cross-validation:** k-fold stratified CV for robust performance estimation.
- **Attention mechanisms:** SE blocks or CBAM for diagnostically relevant feature focus.
- **Mask-guided training:** Leverage BUSI segmentation masks as weak supervision signals.
- **Explainability:** Grad-CAM visualisations to support clinical trust and validation.

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